SALES AND CUSTOMER ANALYTICS SYSTEM

DBMS- Lenskart E-Commerce Database

TASK 1: APPLY AGGREGATE FUNCTIONS

1.1 COUNT() – Total Number of Orders

```
SQL> SELECT COUNT(*) AS Total_Orders
  2  FROM LenskartOrder;

TOTAL_ORDERS
------------
           5
```

1.2 SUM() – Total Sales Amount

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM LenskartOrder;

TOTAL_SALES
-----------
       7400
```

1.3 AVG() – Average Order Amount

```
SQL> SELECT AVG(Total_Amount) AS Average_Order_Amount
  2  FROM LenskartOrder;

AVERAGE_ORDER_AMOUNT
--------------------
               1480
```

1.4 MIN() – Minimum Order Amount

```
SQL> SELECT MIN(Total_Amount) AS Minimum_Order_Amount
  2  FROM LenskartOrder;

MINIMUM_ORDER_AMOUNT
--------------------
                900
```

1.5 MAX() – Maximum Order Amount

```
SQL> SELECT MAX(Total_Amount) AS Maximum_Order_Amount
  2  FROM LenskartOrder;

MAXIMUM_ORDER_AMOUNT
--------------------
               2000
```

TASK 2: GENERATE TOTAL SALES REPORTS

```
SQL> SELECT
  2      COUNT(Order_ID) AS Total_Orders,
  3      SUM(Total_Amount) AS Total_Sales,
  4      AVG(Total_Amount) AS Average_Sales
  5  FROM LenskartOrder;

TOTAL_ORDERS TOTAL_SALES AVERAGE_SALES
------------ ----------- -------------
           5        7400          1480
```

2.2 Sales Report by Order Status

```
SQL> SELECT
  2      Order_Status,
  3      COUNT(Order_ID) AS Total_Orders,
  4      SUM(Total_Amount) AS Total_Sales
  5  FROM LenskartOrder
  6  GROUP BY Order_Status
  7  ORDER BY Total_Sales DESC;
```

```
ORDER_STATUS                 TOTAL_ORDERS TOTAL_SALES
---------------------------- ------------ -----------
Confirmed                               2        3200
Shipped                                 1        1800
Delivered                               1        1500
Processing                              1         900
```

TASK 3: FIND TOP CUSTOMERS BASED ON PURCHASE AMOUNT

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.Customer_Name,
  4      SUM(O.Total_Amount) AS Total_Purchase_Amount
  5  FROM LenskartCustomer C
  6  INNER JOIN LenskartOrder O
  7      ON C.Customer_ID = O.Customer_ID
  8  GROUP BY C.Customer_ID, C.Customer_Name
  9  ORDER BY Total_Purchase_Amount DESC;
```

```
CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
          4
Divya Raj
                2000

          2
Priya Sharma
                1800

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------

          3
Rahul Kumar
                1500

          1
Arun Kumar

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
                1200

          5
Karthik S
                 900
```

TASK 4: IDENTIFY BEST-SELLING PRODUCTS

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      SUM(OI.Quantity) AS Total_Quantity_Sold,
  5      SUM(OI.Quantity * OI.Price) AS Total_Sales
  6  FROM LenskartProduct P
  7  INNER JOIN LenskartOrderItem OI
  8      ON P.Product_ID = OI.Product_ID
  9  GROUP BY P.Product_ID, P.Product_Name
 10  ORDER BY Total_Quantity_Sold DESC;
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
         1
Classic Black Frame
                  1        1200

         2
Blue Light Glasses
                  1        1800

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------

         5
Daily Contact Lenses
                  1         900

         4
Classic Black Sunglasses

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
                  1        2000

         3
Round Metal Frame
                  1        1500
```

TASK 5: PERFORM CATEGORY-WISE SALES ANALYSIS

```
SQL> SELECT
  2      P.Category,
  3      SUM(OI.Quantity) AS Total_Quantity_Sold,
  4      SUM(OI.Quantity * OI.Price) AS Category_Sales,
  5      AVG(OI.Price) AS Average_Product_Price
  6  FROM LenskartProduct P
  7  INNER JOIN LenskartOrderItem OI
  8      ON P.Product_ID = OI.Product_ID
  9  GROUP BY P.Category
 10  ORDER BY Category_Sales DESC;
```

```
CATEGORY                                           TOTAL_QUANTITY_SOLD
-------------------------------------------------- -------------------
CATEGORY_SALES AVERAGE_PRODUCT_PRICE
-------------- ---------------------
Eyeglasses                                                           3
          4500                  1500

Sunglasses                                                           1
          2000                  2000

Contact Lenses                                                       1
           900                   900
```